CHAPTER 24 · DERIVATIONS

WAVE OPTICS

7 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 OPTICAL PATH LENGTH

SYLLABUS 24.2

RESULT optical path = nd

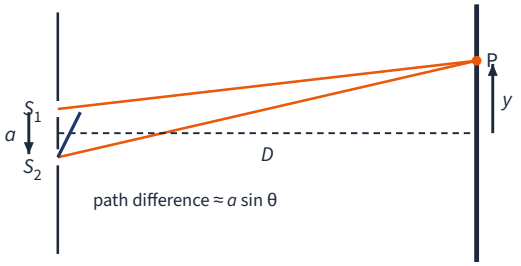
1. Light travels a distance d in a medium of refractive index n at speed $v = \frac{c}{n}$.
2. Time taken in the medium: $t = \frac{d}{v} = \frac{nd}{c}$.
3. In the same time t, light in vacuum travels $ct = c\frac{nd}{c}$.
4. **Optical path = nd**: the equivalent distance in vacuum. Optical path difference = $n_2d_2 - n_1d_1$.

2 FRINGE SEPARATION IN YOUNG'S EXPERIMENT

SYLLABUS 24.3(B)

RESULT $x = \frac{\lambda D}{a}$

1. Slits S_1 and S_2 a apart; screen at distance D ($D \gg a$). Point P is at height y above the centre.
2. Path difference $S_2P - S_1P \approx a \sin\theta \approx a \tan\theta = \frac{ay}{D}$ (small θ).
3. Bright fringe: $\frac{ay}{D} = m\lambda$, so $y_m = \frac{m\lambda D}{a}$.
4. Fringe separation $\mathbf{x = y_{m+1} - y_m = \frac{\lambda D}{a}}$.



3 THIN FILM INTERFERENCE

SYLLABUS 24.4

RESULT $2nt = (m + \frac{1}{2})\lambda$ bright, $2nt = m\lambda$ dark

1. Light falls normally on a thin film of thickness t and refractive index n in air. Ray 1 reflects at the top surface; ray 2 reflects at the bottom surface.
2. Optical path difference between the two reflected rays = 2nt.
3. Phase change: ray 1 reflects from a denser medium (air → film), so it has a phase change of π (equivalent to $\frac{\lambda}{2}$); ray 2 reflects from a less dense medium (film → air), so no phase change.
4. Constructive (bright): $2nt + \frac{\lambda}{2} = (m + 1)\lambda$, so $\mathbf{2nt = (m + \frac{1}{2})\lambda}$.
5. Destructive (dark): $2nt + \frac{\lambda}{2} = (m + \frac{1}{2})\lambda$, so $\mathbf{2nt = m\lambda}$ ($m = 0, 1, 2, \ldots$).

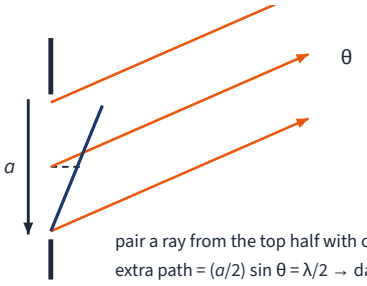
If both reflections have a phase change (e.g. a coating with n between air and glass), the two conditions swap: $2nt = m\lambda$ bright, $2nt = (m + \frac{1}{2})\lambda$ dark.

4 FIRST MINIMUM OF A SINGLE SLIT

SYLLABUS 24.5(B)

RESULT $\sin\theta = \frac{\lambda}{a}$

1. Divide the slit of width a into two halves.
2. Every point in the top half pairs with a point $\frac{a}{2}$ below it. In direction θ the path difference of each pair is $\frac{a}{2}\sin\theta$.
3. If $\frac{a}{2}\sin\theta = \frac{\lambda}{2}$, each pair cancels and the screen is dark.
4. **$\sin\theta = \frac{\lambda}{a}$** for the first minimum.



pair a ray from the top half with one $a/2$ below it
extra path = $(a/2)\sin\theta = \lambda/2$ → dark

5 DIFFRACTION GRATING

SYLLABUS 24.6(B)

RESULT $d\sin\theta = m\lambda$

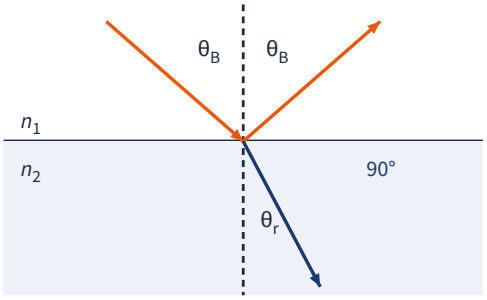
1. Grating spacing $d = \frac{1}{N}$ (N lines per metre).
2. In direction θ, light from adjacent slits has path difference $d\sin\theta$.
3. Bright maxima when all slits are in phase: **$d\sin\theta = m\lambda$** ($m = 0, 1, 2, \ldots$), with m less than $\frac{d}{\lambda}$.

6 BREWSTER'S LAW

SYLLABUS 24.7(F)

RESULT $\tan\theta_B = \frac{n_2}{n_1}$

1. At the Brewster angle the reflected light is completely plane polarised and the reflected and refracted rays are at 90°.
2. $\theta_B + \theta_r = 90°$.
3. Snell's law: $n_1\sin\theta_B = n_2\sin\theta_r = n_2\sin(90° - \theta_B) = n_2\cos\theta_B$.
4. **$\tan\theta_B = \frac{n_2}{n_1}$**.



7 MALUS'S LAW

SYLLABUS 24.7

RESULT $I = I_0 \cos^2\theta$

1. Plane-polarised light of amplitude A_0 (intensity I_0) falls on an analyser whose transmission axis makes angle θ with the plane of polarisation.
2. Resolve the amplitude: only the component parallel to the transmission axis passes, so the amplitude after the analyser is $A = A_0 \cos\theta$.
3. Intensity is directly proportional to the square of the amplitude: $\frac{I}{I_0} = \frac{A^2}{A_0{}^2} = \cos^2\theta$.
4. **$I = I_0 \cos^2\theta$**.